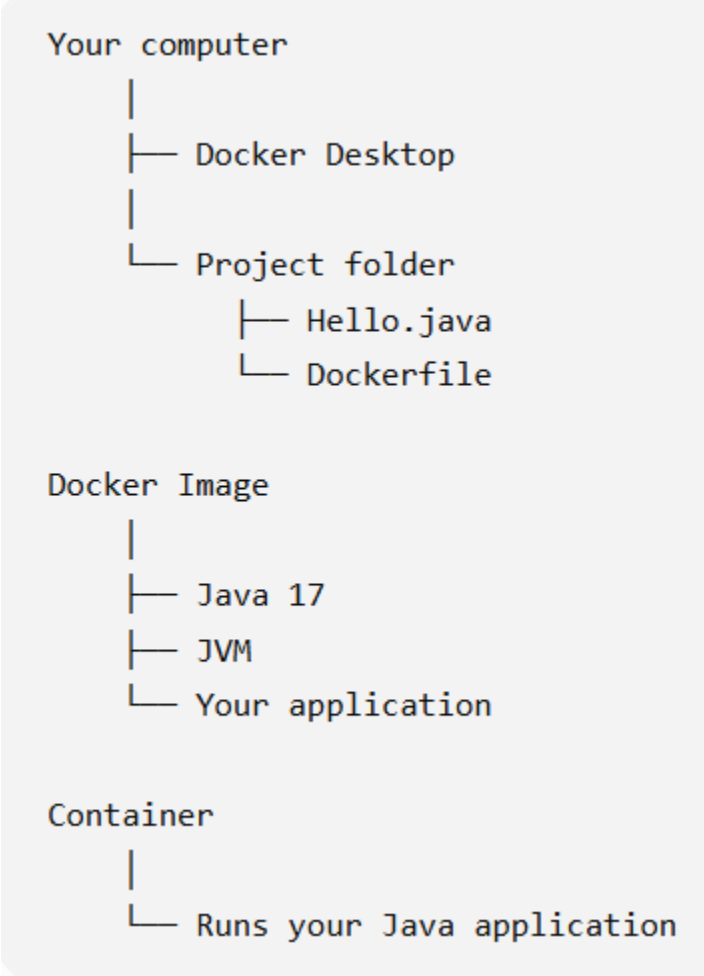
Containerization of a Java 17 Application Using Docker

Aim

To install Docker, create a simple Java 17 application, build a Docker image containing the Java runtime and application, create a container from the image, and execute the application inside the container.

Procedure



Install Docker Desktop

Go to the official Docker website and download **Docker Desktop for Windows**.

Docker Desktop for Windows

Install Docker Desktop using the normal installation procedure.

During installation, keep the recommended **WSL 2** option enabled if Docker Desktop offers it.

After installation, restart Windows if requested.

Start Docker Desktop

Open:

Docker Desktop

Wait until Docker reports that it is running.

Now open:

Command Prompt

or:

PowerShell

Verify Docker Installation

Run: docker --version

You should get something similar to:

Docker version 28.x.x

The exact version may be different.

Now run: docker run hello-world

Docker may download the `hello-world` image.

You should eventually see:**Hello from Docker!**

This confirms that Docker is working.

Create Our Java Project

```
C:\DockerJava
```

```
DockerJava
│
├── Hello.java
└── Dockerfile
```

```
Hello.java
```

```java
public class Hello {

    public static void main(String[] args) {

        System.out.println("Hello from Java 17!");
        System.out.println("Java application is running inside Docker.");

    }
}
```

Dockerfile (There should be no `.txt` extension.)

```
FROM eclipse-temurin:17

WORKDIR /app

COPY Hello.java .

RUN javac Hello.java

CMD ["java", "Hello"]
```

Open Terminal in the Project Folder

Open PowerShell.

Move into the project:

cd C:\DockerJava

Check the files:

dir

You should see:

Dockerfile

Hello.java

Build the Docker Image

Run:**docker build -t java17-app .**

Check the Image

Run:docker images

```
REPOSITORY    TAG       IMAGE ID       SIZE
java17-app    latest    xxxxxxxxxxxx   ...
```

Create and Run a Container

Run:

docker run --name java17-container java17-app

```
Hello from Java 17!
Java application is running inside Docker.
```

Check Containers

Run:

docker ps -a

```
CONTAINER ID   IMAGE         NAMES
xxxxxxxxxxxx   java17-app    java17-container
```

Run the Same Image Again

You don't need to build another image.

Run:

docker run --name java17-container2 java17-app

Output:

Hello from Java 17!

Java application is running inside Docker.

View output again

docker logs java17-container

Stop a Running Container

If a container is currently running:

docker stop java17-container

Remove a Container

After stopping it:

docker rm java17-container

Remove the Image

If you want to remove the image:

```
docker rmi java17-app
```

Result

Thus, Docker Desktop was installed and verified successfully. A Docker image containing Java 17 and the Java application was created using a Dockerfile. A container was created from the image, and the Java application was successfully executed inside the container, producing the expected output.